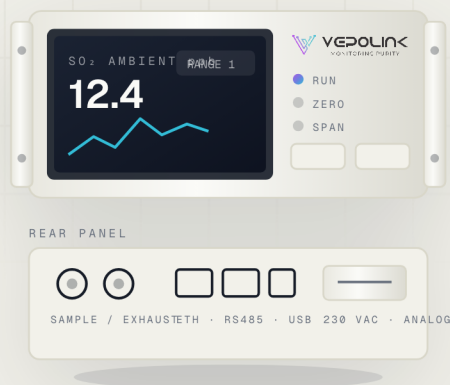


CONTINUOUS AMBIENT AIR QUALITY MONITORING

Ambient SO₂ Gas Analyzer *UV Fluorescence*

Sulphur dioxide measured down to parts per billion by UV fluorescence — with a hydrocarbon kicker, automatic zero correction and a year of logged data on board.

**0.1 · 0.5 · 1** ppm

USER-SELECTABLE RANGES

7 inch

TFT LCD TOUCHSCREEN

1 year

ON-BOARD DATA LOGGING

0.5_{ppb}LOWER DETECTABLE
LIMIT**3**SELECTABLE
MEASURING
RANGES**60**_sTYPICAL RESPONSE
TIME T90**32**_{bit}ARM PROCESSOR,
LINUX BASED

01 OVERVIEW

Sulphur dioxide, *to the last ppb.*

A continuous ambient sulphur dioxide analyzer built on the UV fluorescence method — the accepted technique for regulatory SO₂ measurement. A long-life UV lamp with low sensitivity drift, a built-in kicker to remove hydrocarbon interference and specially coated sample paths give a stable, low-noise reading at ambient concentrations.

The instrument runs a 32-bit ARM processor on a Linux kernel with a 7-inch touchscreen interface, an automatic zero correction system for unattended stability, and Ethernet, RS-232, RS-485, USB and analog interfaces so the analyzer reports into a station, a SCADA system and a regulatory server at the same time.

REPORTED CHANNELS

- SO₂ INSTANT
- 1-MIN AVERAGE
- HOURLY AVERAGE
- 24-HR AVERAGE
- SAMPLE FLOW
- CHAMBER TEMP
- CHAMBER PRESSURE
- LAMP INTENSITY
- ZERO & SPAN STATUS
- FAULT DIAGNOSTICS



Stable UV Source

A UV lamp selected for low sensitivity drift keeps span stable between calibrations and extends service intervals.



Built-In Hydrocarbon Kicker

Interfering hydrocarbons are removed ahead of the chamber, so the fluorescence signal reflects SO₂ alone.



Touchscreen Operation

A 7-inch TFT LCD touch interface puts live values, trends, ranges and diagnostics in front of the operator.



One Year of Data On Board

An internal NAND-flash logger holds up to a year of one-minute averages, downloadable over USB or network.

02 MEASUREMENT PRINCIPLE

How UV fluorescence *counts* SO₂.

Sulphur dioxide absorbs ultraviolet light and re-emits it at a longer wavelength. The intensity of that fluorescence is proportional to concentration — a direct optical measurement with no reagents and no consumables in the sample path.

UV FLUORESCENCE

Excite, Emit, CountUV LAMP · REACTION CHAMBER ·
PHOTOMULTIPLIER

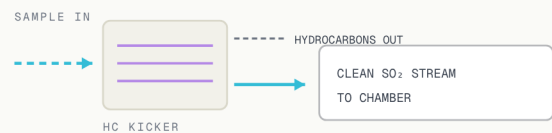
Sample air passes a chamber illuminated by ultraviolet light. Excited SO₂ molecules fluoresce as they relax; a photomultiplier counts that emission and the analyzer converts intensity to concentration.

Signal $\propto$ SO₂ concentration

INTERFERENCE REMOVAL

Hydrocarbon Kicker

SELECTIVE PERMEATION AHEAD OF CHAMBER



Aromatic hydrocarbons fluoresce in the same region and would inflate the reading. An in-built kicker strips them from the sample stream while SO₂ passes through untouched.

Clean SO₂ signal

STABILITY

Auto Zero & Compensation

AUTOMATIC BASELINE · T AND P CORRECTION



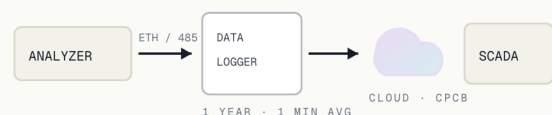
An automatic zero correction cycle re-establishes the baseline without operator intervention, while temperature and pressure compensation holds the reading steady as ambient conditions move through the day.

Unattended long-term operation

DATA PATH

Logging & Reporting

ETHERNET · RS-232 / 485 · USB · ANALOG



Readings are stored internally as one-minute averages for up to a year and transmitted live to a station data logger, plant SCADA or a pollution control board server.

Download over USB or network

03 SAMPLE PATH

From inlet to *reading*.

Everything between the sample line and the display is designed to keep the molecule you are measuring intact and everything else out of the signal.

STEP 01 Filtered intake Ambient air is drawn through a particulate filter and inert sample line into the analyzer at a controlled flow.	STEP 02 Kicker conditioning The hydrocarbon kicker removes aromatic compounds that would otherwise fluoresce and bias the measurement.	STEP 03 Coated chamber Reaction chamber and sample tubing carry a special coating that prevents SO ₂ adhesion, improving detection limit and precision.	STEP 04 Detect and report Fluorescence is counted, corrected for temperature and pressure, then displayed, logged and transmitted.
---	--	--	--

WHY COATING MATTERS

SO₂ is a sticky molecule. On untreated surfaces it adsorbs on the way in and desorbs later, smearing fast changes and raising the effective detection limit. Coated chamber walls and sample tubing keep the molecule in the gas phase, which is what makes single-digit ppb measurement repeatable rather than theoretical.

RANGE SELECTION

Three ranges — 0–0.1, 0–0.5 and 0–1 ppm — are user selectable from the touchscreen. Urban background sites usually run on the lowest range for resolution; industrial fence-line and near-source installations use a wider range so episodes are captured without saturating.

WHY SULPHUR DIOXIDE IS MONITORED

SO₂ is a colourless, pungent gas released mainly by fossil fuel combustion and industrial process emissions, with volcanic activity as the natural source. At elevated concentrations it aggravates respiratory conditions and drives the formation of acid rain, which acidifies soil and water and damages ecosystems and structures. Continuous measurement is what turns that risk into something a plant, a city or a regulator can act on.

04 SPECIFICATIONS

Technical *specifications.*

MEASUREMENT

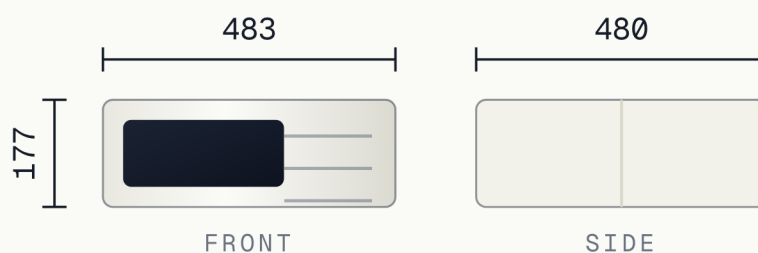
Parameter	Ambient sulphur dioxide, SO ₂
Working Principle	UV fluorescence
Measuring Ranges	0 - 0.1 / 0.5 / 1 ppm, user selectable
Lower Detectable Limit	0.5 ppb
Precision	±1 % of reading
Linearity	±1 % of full scale
Zero Drift	< ±1 ppb per 24 hours
Span Drift	< ±1 % FS per 7 days
Response Time	T90 typically 60 s
Interference Removal	Built-in hydrocarbon kicker
Sample Path	Coated chamber and tubing to prevent adhesion

SYSTEM · COMMS · BUILD

Display	7-inch TFT LCD with touch interface
Controller	32-bit ARM Cortex-M4, Linux based multiprocessing
Zero Correction	Automatic · internal zero optional
Compensation	Temperature and pressure
Data Logger	NAND flash · up to 1 year of 1-minute averages
Interfaces	Ethernet · RS-232 · RS-485 · USB · analog out
Data Download	USB export or network transfer
Power Supply	230 VAC · 50 Hz
Mounting	19-inch rack or bench, station cabinet
Operating Conditions	5 - 40 °C, non-condensing
Enclosure	Weather-protected station cabinet for outdoor sites

INSTRUMENT DIMENSIONS

INDICATIVE FRONT & SIDE ELEVATION · MM



05 INSTALLATION & UPKEEP

Racked, calibrated, *left running*.

The analyzer is installed in a station cabinet with the sample manifold, calibration source and data logger. Per CPCB practice, the sample inlet sits in free-flowing ambient air **3 to 10 m above ground**, with an inert, short sample line and a scheduled **zero and span check** traceable to certified gas.

01 · MOUNTING 19-inch rack or bench Standard rack fitment alongside other CAAQMS analyzers, sharing the station's manifold, power and data logger.	02 · SAMPLE LINE Inert and short PTFE line with a particulate filter, kept short and unheated-clean so SO₂ reaches the chamber without loss.	03 · ZERO AIR Clean baseline source A zero air unit supplies scrubbed air for the automatic zero cycle and for periodic manual verification.
04 · CALIBRATION Multipoint traceable Span with certified SO₂ from a permeation source or cylinder via a multipoint calibrator, on the site's audit schedule.	05 · CONSUMABLES Filter and lamp Routine upkeep is the inlet filter, periodic leak check and lamp inspection. No reagents and no wet chemistry.	06 · DIAGNOSTICS Reported with the data Lamp intensity, chamber temperature and pressure, flow and zero/span status travel alongside every reading.

PART OF A FULL STATION

The SO₂ analyzer is normally specified with NO_x, CO and ozone analyzers, a multipoint calibrator and zero air unit, a particulate monitor and a meteorological mast. Sharing a time base, cabinet and data path, they report as one station record — which is what a pollution control board expects to receive.

COMMISSIONING & AUDIT

Commissioning covers leak and flow verification, a multipoint linearity check across the working range, and a documented zero and span baseline that becomes the reference for subsequent audits. Thereafter the automatic zero cycle carries day-to-day stability and the scheduled span check confirms it against certified gas, so the record stays defensible without a technician on site.

06 APPLICATIONS

Where it *works*.

Wherever sulphur is burned or processed, SO₂ has to be measured — for compliance, for public health reporting, and for understanding what a source actually contributes downwind.

 Regulatory CAAQMS Networks	 Thermal Power Stations	 Refineries & Petrochemicals
 Chemical & Fertiliser Plants	 Smelters & Metallurgy	 Urban & Traffic Corridors
 Ports & Marine Fuel Zones	 EIA & Baseline Studies	 Research & Air Quality Studies

WHAT THE DATA SUPPORTS

Continuous SO₂ records serve three audiences at once: regulators, who need hourly and daily values against national standards; operations, who need to see the effect of fuel switching, scrubber performance and load changes; and communities, who need a credible answer about the air at the fence line. One measurement, three conversations.

SITING NEAR A SOURCE

Near-source installations are usually placed downwind of the prevailing direction with a second station upwind as a background reference, so the contribution of the site can be separated from what arrives at the boundary already. Where fuel sulphur or load varies, a wider measuring range keeps episodes on scale rather than clipped.

Frequently asked *questions*.

Q What is the working principle?

UV fluorescence. Sulphur dioxide excited by ultraviolet light emits fluorescence in proportion to its concentration, which a photomultiplier detects and the analyzer reports in ppb or ppm.

Q Which parameter does it measure?

Ambient sulphur dioxide concentration, for continuous ambient air quality monitoring in regulatory, industrial and research applications.

Q Is it suitable for continuous monitoring?

Yes. It is designed for real-time, uninterrupted operation, with automatic zero correction so the baseline holds without an operator present.

Q Does it meet CPCB requirements?

The analyzer is designed to meet CPCB and regulatory ambient air quality monitoring guidelines, and reports in the formats a state or central board server expects.

Q Can it be installed outdoors?

Yes, housed in a weather-protected station cabinet. The analyzer itself is a rack instrument; the cabinet provides the environmental protection and shared services.

Q How is hydrocarbon interference handled?

A built-in kicker removes aromatic hydrocarbons upstream of the reaction chamber, because those compounds fluoresce in the same region and would otherwise read as SO₂.

Q What ranges are available?

0–0.1, 0–0.5 and 0–1 ppm, selectable by the user from the touchscreen to suit background, urban or near-source conditions.

Q How much data does it hold?

The internal NAND-flash logger stores up to one year of one-minute averaged data, exportable over USB or pulled across the network.

Q How does it integrate with a station?

Ethernet, RS-232, RS-485 and analog outputs let the analyzer report to a station data logger, plant SCADA and a regulatory server simultaneously.

Q What maintenance does it need?

Inlet filter changes, periodic leak and flow checks, and scheduled zero and span verification with certified gas. There are no reagents or wet chemistry.

This brochure is general product information. Configuration, dimensions and performance figures are indicative and confirmed at order. For installation guidelines and full technical detail, refer to the user manual supplied with the product. Specifications may change without notice as the product is improved.

SULPHUR DIOXIDE, MEASURED CONTINUOUSLY

Know your SO₂. *In real time.*

Tell us the site, the range you expect and who receives the data, and we'll configure the analyzer and the station around it — calibration, zero air, cabinet and reporting. Pick how you'd like to start.

Talk to our engineers

01

Speak directly to the team that specifies, installs and calibrates ambient gas analyzers. No call centre.

+91 99538 86178



Request a station configuration

02

RECOMMENDED

Send your site and reporting requirement — we'll spec ranges, calibration, zero air, cabinet and communications.

sales@vepolink.com



Book a demonstration

03

See a running analyzer and the reporting dashboard with our engineering team.

Request a demo

